Customer Churn Analysis and Customer Intelligence

Portfolio Project Report

Objective: analyze customer cancellations, compare churn across segments, and recommend retention actions using SQL and Python.

Methodology

The project queries three SQLite tables, standardizes columns, parses date and numeric values, aggregates support interactions at customer level, and joins the tables. It engineers churn status, tenure, churn month, risk tier, and revenue-at-risk fields before calculating segment KPIs and visualizing patterns.

KPI Summary

Metric	Value
Total customers	21
Churned customers	6
Churn rate	28.57%
Retention rate	71.43%
Average tenure	1028.7 days
Customers with complaints	7
Customers with escalations	0

Recommendations

Prioritize high-risk active customers, investigate cancellation reasons and escalated support cases, and compare churn by contract and plan before changing pricing or packaging. Monitor the same KPIs after retention interventions.

Limitations

The supplied sample contains 21 customer/subscription records and 9 support interactions. Results are illustrative and should not be generalized without a larger, recent dataset. Tenure for active customers uses 2024-09-30 as the snapshot date. Confirm units/currency before communicating revenue impact.